🕵️ SOC INVESTIGATION REPORT

Brute Force Login Detection (Linux System – Splunk SIEM)

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1. Executive Summary

This report documents the investigation of suspicious SSH authentication activity detected on a Linux system. Using Splunk SIEM and Linux secure logs, multiple failed login attempts were identified, indicating a potential brute-force attack targeting user accounts.

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2. Incident Description

A high volume of authentication attempts, including failed and successful SSH logins, was observed from a specific source IP address (10.10.242.248). The pattern of repeated login failures, combined with attempts against multiple usernames, suggests automated brute-force activity.

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3. Investigation Methodology

The investigation was conducted using Splunk SIEM by querying Linux authentication logs (linux_secure). The following steps were performed:

- Retrieved SSH authentication events including successful and failed logins
- Identified repeated failed login attempts from a specific source
- Extracted targeted usernames from log data
- Analysed frequency and duration of login attempts
- Correlated authentication activity to determine attack patterns

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4. Evidence & Findings

4.1 Authentication Activity Analysis

The following query was used to retrieve authentication events:

```
index="linux-alert" sourcetype="linux_secure" 10.10.242.248
| search "Accepted password for" OR "Failed password for" OR "Invalid user"
| sort + _time
```

Findings:

- Multiple **“Failed password”** and **“Invalid user”** entries observed
- Occasional **“Accepted password”** entries (if applicable, update if none)
- Activity occurred over a continuous time period
- Logs indicate repeated authentication attempts from the same source

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4.2 Targeted User Analysis

The following query was used to identify targeted usernames:

```
index="linux-alert" sourcetype="linux_secure" 10.10.242.248 "Failed password for"
| rex field=_raw "Failed password for(?: invalid user)? (?<username>\S+)"
| stats earliest(_time) as start latest(_time) as end count by username
| eval duration_minutes=round((end-start)/60,0)
| table username count duration_minutes
| sort - count
```

Findings:

- Multiple user accounts were targeted
- Certain usernames experienced significantly higher login attempts
- Attack duration varied per account
- Pattern suggests automated username enumeration followed by brute-force attempts

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5. Analysis

The observed authentication activity shows clear indicators of a brute-force attack:

- High frequency of failed login attempts
- Repeated targeting of specific usernames
- Consistent source IP generating traffic
- Presence of invalid user attempts indicating enumeration

This behavior aligns with automated tools attempting to gain unauthorized access via SSH.

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6. MITRE ATT&CK Mapping

- **T1110 — Brute Force**
- **T1021.004 — Remote Services: SSH**

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7. Risk Assessment

- **Severity:** Medium (High if successful login confirmed)
- **Impact:** Potential unauthorized system access
- **Likelihood:** High (active attack behavior observed)

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8. Recommendations

- Block or blacklist the malicious IP address (10.10.242.248)
- Disable SSH password authentication (use key-based authentication)
- Disable root login via SSH
- Implement fail2ban or intrusion prevention tools
- Enforce strong password policies
- Enable Multi-Factor Authentication (MFA) where applicable
- Continuously monitor authentication logs via SIEM

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9. Conclusion

The investigation confirms that the activity detected in the Linux authentication logs is consistent with a brute-force attack targeting SSH access. Although no confirmed compromise may have occurred, the attack demonstrates a clear attempt to gain unauthorized access and requires immediate mitigation.